Project Title

Ecommerce_Intelligence_Project

Business Problem Statement

Create an end-to-end data analytics pipeline to transform fragmented e-commerce data into a unified intelligence system. The goal of the assignment is to be able to join multiple tables and analyse them to explore findings about an entire e-commerce platform.

Perform structured analysis across the following dimensions:
Customer Analysis

- New vs repeat customers
- High-value vs low-value customers
- Geographic distribution of customers

Revenue and Order Analysis

- Monthly revenue trends
- Order volume trends

- Peak sales periods

Product Analysis

- Top-selling product categories
- Revenue contribution by category
- Product demand distribution

Seller Analysis

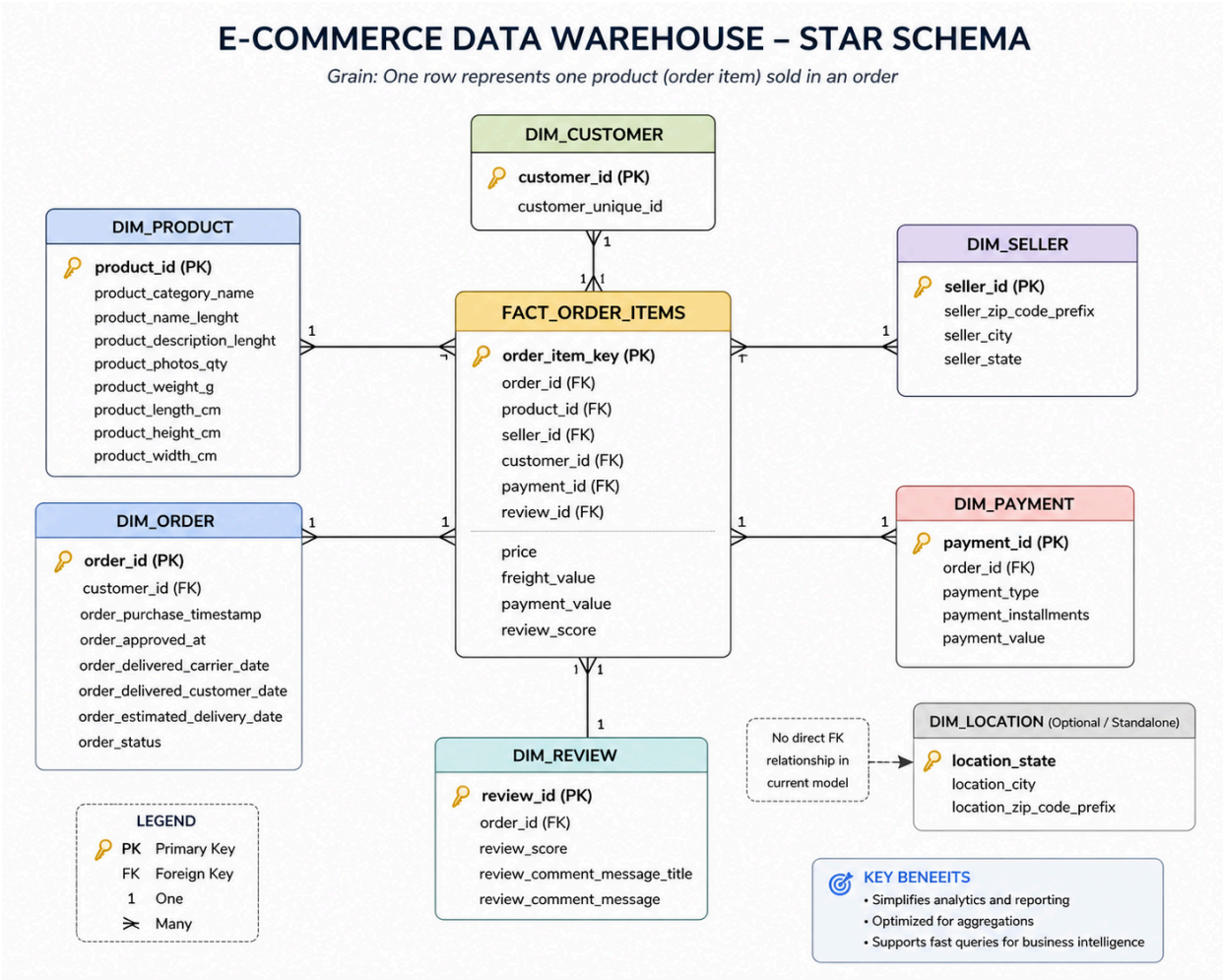
- Top-performing sellers
- Seller contribution to revenue
- Seller distribution

Review and Satisfaction Analysis

- Distribution of review scores
- Relationship between delivery time and ratings
- Identification of dissatisfaction patterns

Dataset Description

Please find the ER Diagram



Methodology

Methodology: Data Integration and Exploratory Analysis

This analysis is based on the integration of multiple relational tables from the Olist e-commerce dataset, transformed into a unified analytical dataset suitable for exploratory data analysis (EDA).

Data Integration Approach

The dataset was constructed by performing structured joins across key transactional and dimensional tables using primary–foreign key relationships. The `orders` table serves as the central entity and is joined with:

- `customers` using `customer_id` to capture customer-level attributes. Analyze `customers` data using `customer_unique_id`
- `order_items` using `order_id` to obtain product, seller, and item-level details
- `payments` using `order_id` to incorporate transaction values
- `reviews` using `order_id` to include customer satisfaction metrics
- `products` and `sellers` via `product_id` and `seller_id` respectively to enrich product and seller dimensions

These joins result in a denormalized dataset at the **order-item level**, enabling granular analysis across multiple dimensions.

Feature Engineering
Additional derived features were created to support analysis:

- Time-based features such as `order_month` and `delivery_time_days`
- Customer segmentation flags (e.g., new vs repeat customers)
- Value-based metrics such as total order value and customer lifetime value
- Satisfaction indicators derived from review scores (e.g., dissatisfied vs satisfied)

Exploratory Analysis Framework

The analysis was conducted across multiple business dimensions:

Customer Analysis
Customers were segmented into new vs repeat buyers based on purchase frequency, and into high-value vs low-value segments using total spend distribution. Geographic patterns were analyzed where location data was available to understand customer concentration.

Revenue and Order Analysis
Temporal aggregation was used to evaluate monthly revenue and order volume trends. Time series analysis helped identify seasonality and peak sales periods.

Product Analysis
Product categories were analyzed based on sales volume and revenue contribution. Demand distribution was evaluated using order frequency across products and categories.

Seller Analysis
Seller performance was assessed by aggregating revenue and order volume at the seller level. Distribution analysis highlighted concentration of revenue among top sellers and the presence of a long-tail seller base.

Review and Satisfaction Analysis
Review score distributions were examined to assess overall customer satisfaction. Delivery time

was correlated with review scores to evaluate service impact on customer experience. Dissatisfaction patterns were identified by analyzing low review scores in relation to delays, sellers, and product categories.

This structured approach enables a comprehensive understanding of customer behavior, operational performance, and key drivers of satisfaction within the e-commerce ecosystem.

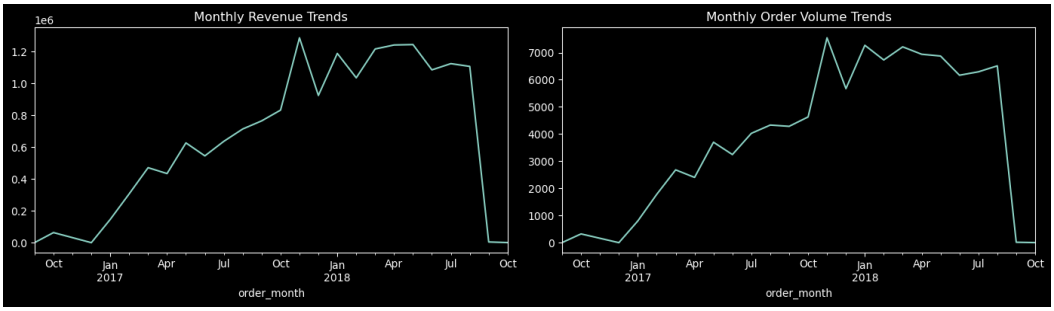
Key Findings

Customer Analysis

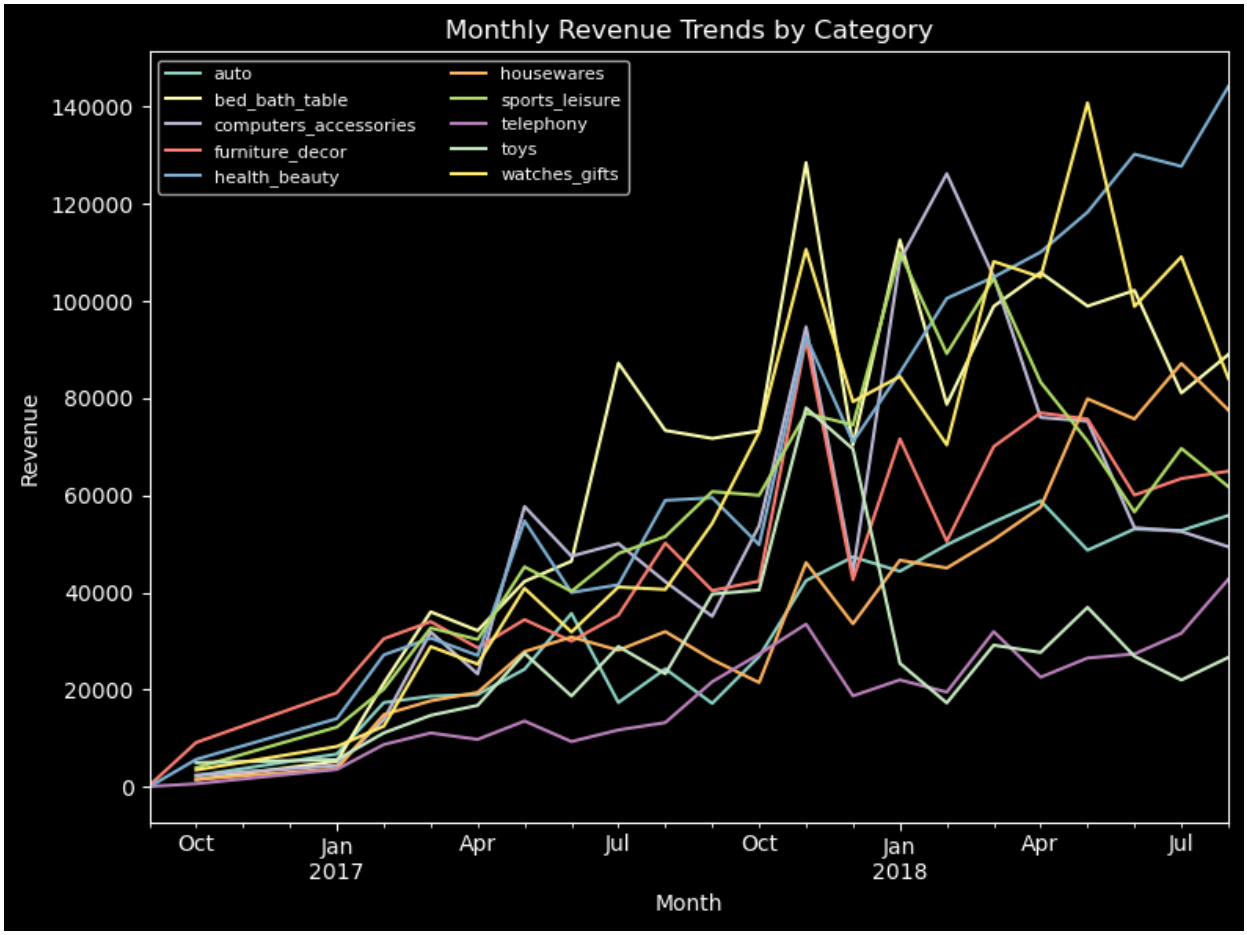
1. New vs Returning Customers: Analyze the proportion of new customers (first-time buyers) versus returning customers. This can help identify customer loyalty and retention rates.
    - <1% of customers are not first time buyers. This means that the majority of the customers are not returning customers. This is a red flag for the business as it indicates that the business is not able to retain customers and build loyalty.
    - The business needs to focus on improving customer retention and loyalty to increase revenue and profitability
2. High-value vs low-value customers
    - High* value customers account for 33.6% of total value whereas they are just 7.9% of the customer base! *(*75% percentile is 177.2; 25th percentile is 62.5. The IQR*1.5 + 177.2 = 350 let's assume above that is high value and below that is low value)*
3. Geographic distribution of customers
    - Top 2 cities sao paulo and rio de janeiro account for 47% of the revenue.
    - Top 3 states **SP, RJ, MG** account for 60%+ of the revenue

Revenue and Order Analysis

1. Monthly revenue trends & Monthly Order Volume trends
    - As seen in the charts below, looks like both revenue and orders follow a very similar pattern
    - Revenue peaked in Oct 2017. We can further analyze the sales by category to see if there are any specific categories that are driving the sales in October.



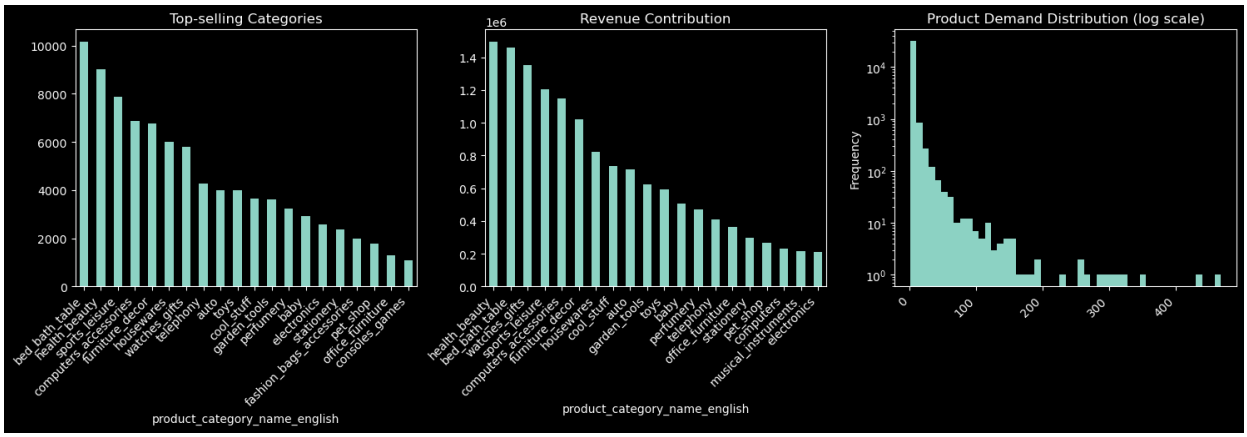
2. Analyzing the category: “Bed bath Table” and “Watches” peaked in Oct 2018



Product Analysis

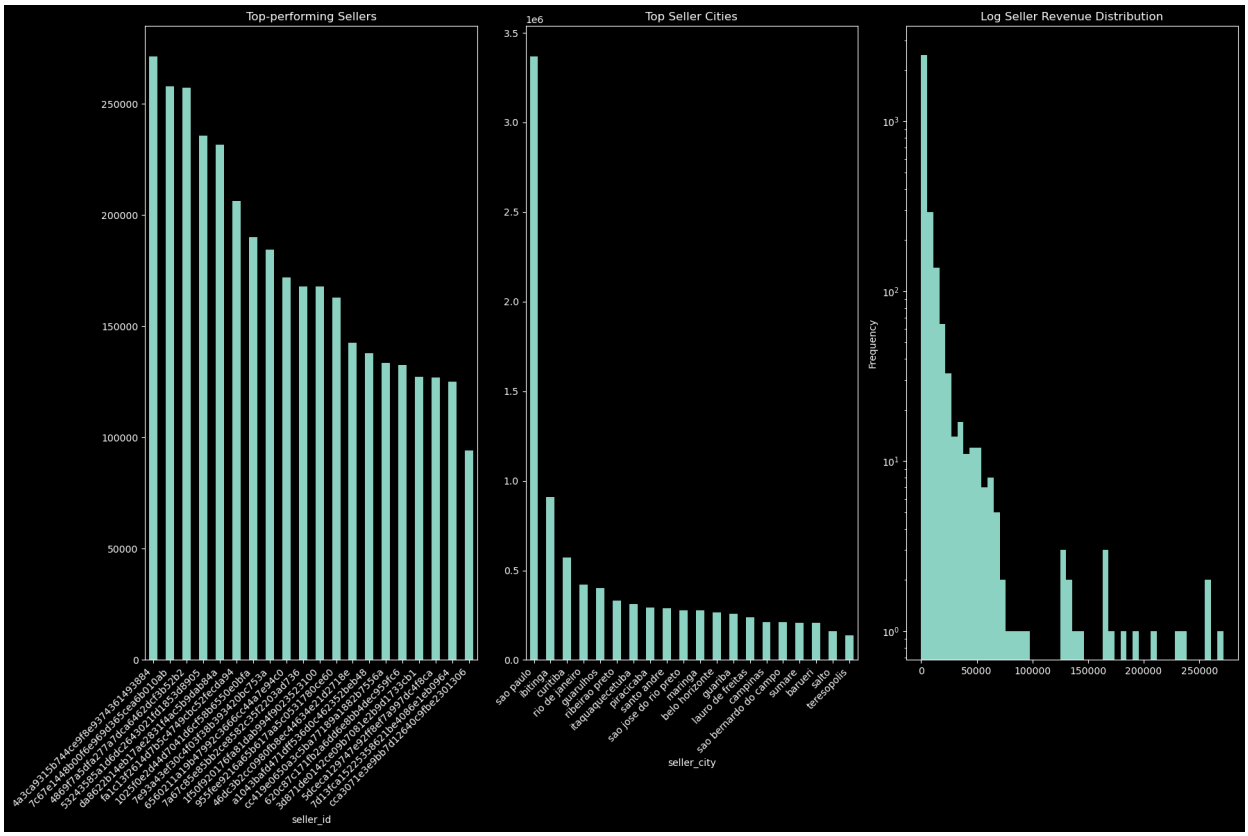
- Top-selling product categories
    - ‘Bed Bath Table’ and ‘Health_Beauty’ are top two
- Revenue contribution by category

    - Health Beauty has the highest revenue contribution even though it has lower quantified sold than ‘Bed bath Table’. These two are the top two categories by selling product categories.
- Product demand distribution
    - Distribution is right skewed, with a long tail



Seller Analysis

- Top-performing sellers
    - Top 20% of sellers contribute 82.1% of the revenue. This indicates a highly skewed distribution of revenue among sellers, with a small percentage of sellers generating the majority of the revenue. This is a common pattern in marketplaces and suggests that the business may want to focus on supporting and retaining these top-performing sellers while also finding ways to help lower-performing sellers improve their sales.
    - Top 10% of sellers contribute 66.2% of the revenue.
- Seller distribution
    - Top 15 sellers each account for a lot of transactions (100k+)
    - Sao Paolo has majority of sellers!



Review and Satisfaction Analysis

- Distribution of review scores
    - Generally, the distribution seems to be 5 heavy with fair bit of values at 1 too.

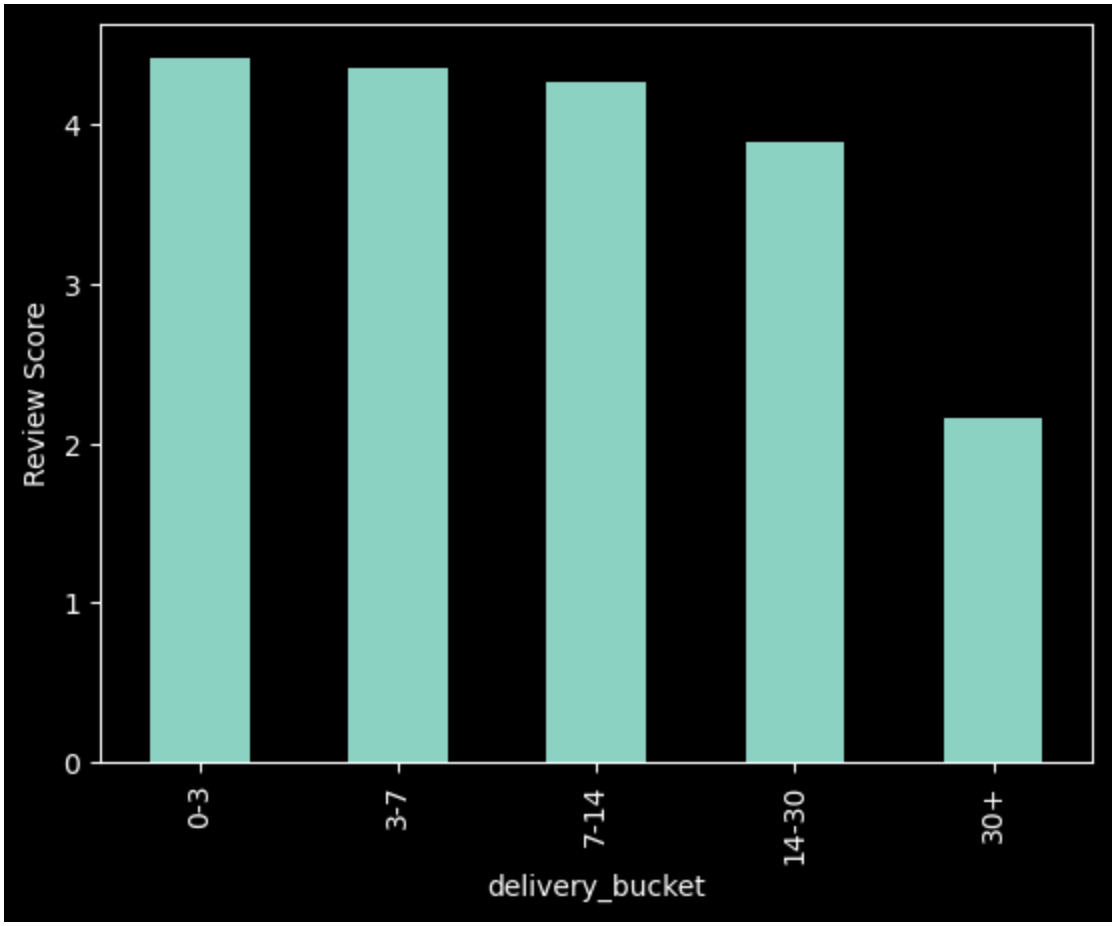
- Relationship between delivery time and ratings
    - The correlation between delivery time and review score is -0.31, which indicates a moderate negative relationship. This suggests that longer delivery times are moderately associated with lower review scores, but the relationship is not strong.
    - Other factors may also be influencing review scores, and delivery time alone does not fully explain customer satisfaction.

Business Insights & Recommendations

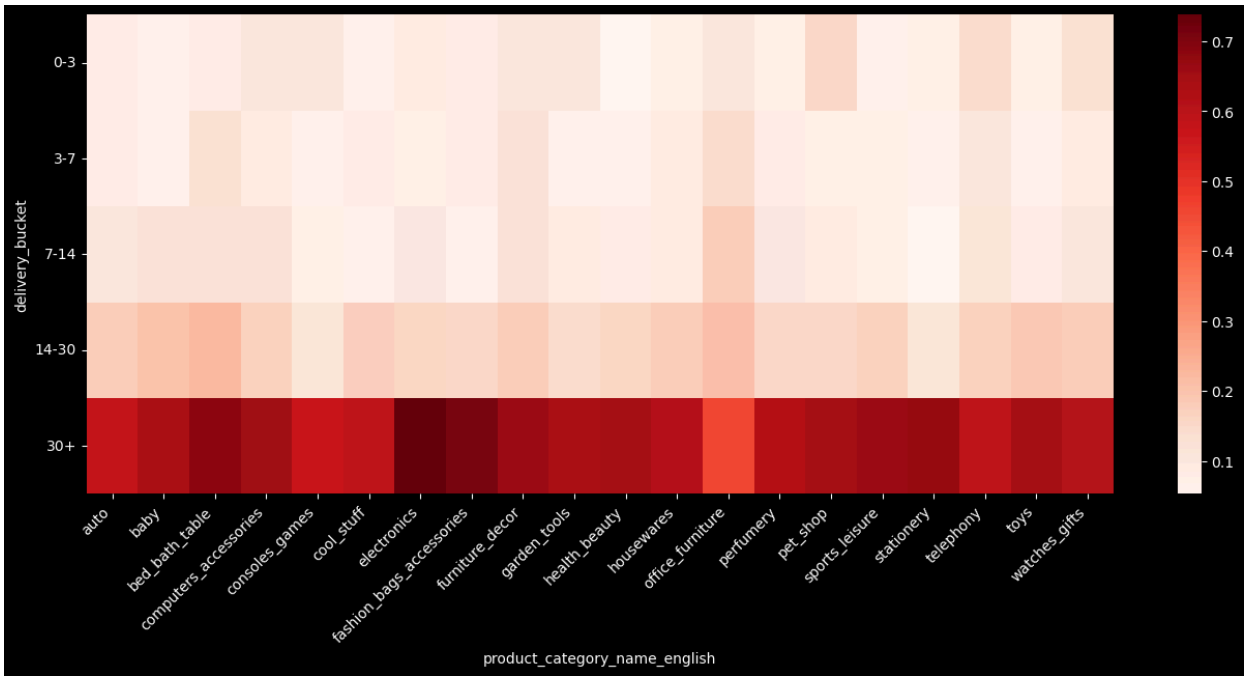
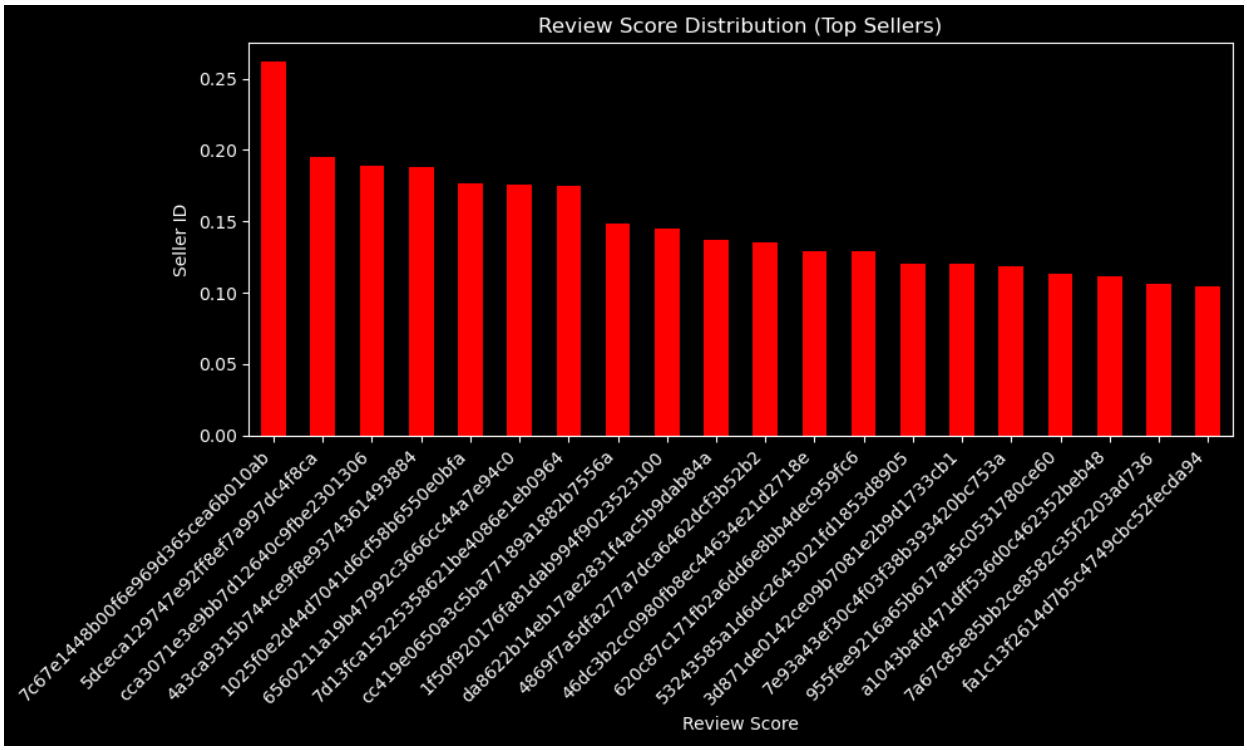
- **Identify top revenue-driving factors**
    - Top 20% of the customers drive 82% of the total revenue
    - Sellers are mostly from 3 states and 2 cities(Sao Paulo & Ibitinga) dominate seller presence
        - **SP** is the top seller state as well as customer state. Focus there as bulk of revenue is form that state
    - Health-Beauty, Bed-Bath-table are the categories that drive most revenue

- **Highlight customer behavior patterns**
    - Most customers are one and done; there is potential to see if this is mostly driven through promotions
    - States SSP, RJ and MG drive most of the customer base (60%+ of revenue)
    - Top 2 cities Sao paulo and Rio de janeiro account for 47% of the revenue

- **Evaluate the operational inefficiencies**
    - **SP** is the top seller state however
    - Review score drops by delivery time. Work on reducing the delivery time for better customer experience



- **Provide strategic recommendations**
    - Customer retention is poor.<1% of customers are not first time buyers. This means that the majority of the customers are not returning customers. This is a red flag for the business as it indicates that the business is not able to retain customers and build loyalty.
        - The business needs to focus on improving customer retention and loyalty to increase revenue and profitability
    - Dis-satisfaction is very high in higher delivery buckets (30+). It would help the marketplace to find ways to reduce delivery times
        - Amongts 30+ days bucket, Electronics and Bed-bath-table category especially gets hit by dis-satisfied users the most



- Seller (7c67….) has the highest dissatisfied users and it might be good to check on their product and user experience. Are customers really suffering?